Title: S8 MISCELLANEOUS TOPOLOGICAL DESIGN RULES (MTDR)

1. PURPOSE/SCOPE

1.1 Purpose

Document the Latch-up and ESD Design Rules for products using the S8 process.

1.2 Scope

This specification is intended for use by ESD Technology and Design Engineers in development of the new products using the S8 process.

Dimensions included in the spec are final = drawn.

2. RESPONSIBILITIES

2.1 Spec ownership

This spec is to be defined and revised by ESD Technology engineers only. If other parties need to change any part of this spec, the revision needs to be reviewed and agreed upon by ESD Technology engineers.

2.2 Performing the Process

ESD Technology and Design Engineers are responsible for following this spec.

2.3 Record keeping

All changes need to be documented in detail using History page.

3. REFERENCED DOCUMENTS

3.1 Referenced Documents (DMS)
CTRL-Click (click if PDF file) on the above link to access the Document Management System (DMS) details page for the spec. Then click on the "Reference Documents" button to bring up the list of referenced documents.

4. MATERIALS: N/A

5. EQUIPMENT: N/A

6. SAFETY: N/A

7. CRITICAL REQUIREMENTS SUMMARY

7.1 Latchup endurance requirement

Refer to spec # 01-00081

7.2 ESD endurance requirement, reference spec # 25-00020

7.2.1 ESD pass level on all pins using Human Body Model (HBM)

7.2.2 HBM test standards

7.2.3 ESD pass level using the Charged Device Model (CDM)

7.2.4 CDM test standards

8. OPERATING PROCEDURES AND RESPONSIBILITIES

8.1 Latchup Generic Rules

8.2 SONOS-Specific Latchup Rules

8.3 Super High Voltage Latchup Rules – section deleted

8.4 Signal Pad Latchup Rules

8.5 Latchup Special Cases

8.6 Signal Pad Main ESD Protection

8.7 Signal Pad Secondary ESD Protection

8.8 Power Clamping Rules

8.9 Bus Resistance Rules

8.10 Main ESD Clamp Hookup

8.11 Miscellaneous Rules

9. QUALITY REQUIREMENTS

9.1 Design engineers and customers are responsible for providing design databases that are clean to the rules included in this specification.

9.2 Not all rules in this spec are checked automatically and must be checked manually. All rules and waivers must be signed off by customer.

10. RECORDS

10.1 All records shall be maintained as specified in 00-00064 - Record Retention Policy and Procedures.

10.2 Document History
Document History Query
To access the document revision history, CTRL-click (click if PDF document) on the above link to bring up the web query page and enter the spec number in the dialog box and click on submit.

11. PREVENTIVE MAINTENANCE

11.1 This specification must be updated along with the objective specification change.

12. POSTING SHEETS/FORMS/APPENDIX

12.1 Appendix A identifies the library & .ddc where the approved ESD cells are located.

12.2 Appendix B contains the LU DRC form description

	Definitions and Terminology
x.1	"areaid.ed" surrounds any diffusion or ESD nwell tap connected to a signal pad. Note: algorithm does not flag devices with areaid:en that is drain extended devices.
x.2	*ESD_diffusion* is defined as any diffusion or ESD_nwell_tap connected metallically or through a resistor to a pad or to Vss/Vcc that is covered by areaid.ed. Note: CAD flow generates layers ESD_PSD and ESD_NSD in VLW to identify diff metallically connected to I/O Pad. It also generates layer io_res to identify ESD resistors.
x.3	*ESD_nwell_tap* is defined as n+ tap coincident with nwell such that n+ tap and nwell are completely surrounded by and abutting n+ diff on all edges, within areaid:ed.
x.4	*ESD_diode* is defined as any nwell (other than any ESD nwell tap) covered by areaid:ed and areaid:de that does not contain poly.
x.5	*esd_diff_tap_nwell* is defined as ESDID AND diff_tap_nwell.
x.6	*diff_tap_nwell* is defined as tap_nwell INSIDE diff_hole.
x.7	*tap_nwell* is defined as tap INSIDE nwell.
x.8	*diff_hole* is defined as Hole(diff).
x.9	*ESD_NFET* is defined as (any Ndiff covered by areaid:ed abutting ESD_nwell_tap) Or (any Ndiff covered by areaid:ed abutting gate within 3.5um of ESD_nwell_tap) Or (any Ndiff abutting ESD_nwell_tap within areaid.ed) a double tap guardrings.
x.10	*ESD_FET* is defined as (any Pdiff covered by areaid:ed within a double tap guardrings) Or ESD_NFET
x.11	*met_ESD_resistor* is defined as Metal resistor inside areaid:ed.
x.12	*I/O_or_Output_Pmos* is defined as ESD P+ diffusion overlapping poly and overlapping ESD source/drain diffusion connected to I/O or output net.
x.13	*flare_gates* is defined as 45 degrees poly, straddling diff inside areaid:ed.
x.14	*Non_Vcc_nwell* is defined as any nwell or Dnwell connected to a positive power supply through $>$ 10Ω, or connected to Vss or connected to a signal pad.
x.14a	*At RISK Non_Vcc_nwell* is any nwell containing p+ diffusion connected to a positive external power supply ≥ 3.3V, where the nwell is not metallically connected to the same positive power supply.
x.15	*"RF Pins"* that require special ESD protection schemes must be reviewed by the ESD Review Board and the pin labels must use the prefix "RF_" for pin identification. The proper label format is "RF_userPinName".
x.16	*shorted_on_die* : Two power (or ground) nets are shorted on die if they are connected on the die through metal buses.
x.17	*shorted_thru_package* : Two power (ground) nets are shorted through package if they are not connected on the die, but connected through bond wires or package ground planes.
x.18	*isolated_on_die* : Two power (ground) nets are isolated_on_die if they are not shorted on die, even if they are shorted through package.
x.19	*internal Vcci, Vssi* : Internal power and ground supplies (Vcci or Vssi) are supplies with highest capacitance on die and usually connect to the memory array and the majority of internal circuitry. All other *isolated_on_die* supplies are labeled as Vccn or Vssn, where n=1,2…
x.20	*pwell* : is defined as deep nwell not nwell.

	Definitions and Terminology
x.21	*Positive power supply:* Any external or regulated positive power supply (Vcc net)
x.22	“areaid.ly” encloses any circuitry meeting the loose tapping rule but violating the tight tapping rule. This part of the circuitry must be placed >50um away from signal pad injection sources at the chip-level.
x.23	“areaid.inj” encloses any circuitry deemed sensitive (by design team) to injected substrate carriers. This part of the circuitry must be placed >100um away from signal pad injection sources at the chip-level. areaid.inj encloses any PVT compliant circuitry
x.24	Sensitive Circuitry: Circuitry that is susceptible to disturb issues associated with injected substrate carriers from signal pad connected diffusions/well. The design community must assess their design’s sensitivity to this phenomenon.
x.25	Resistive voltage divider: A resistive connection between power and ground nets, used to produce a fraction of the power supply voltage at its output. It can be composed of any combination of poly, ESD poly, pwell, diffusion, ESD diffusion, and ESD metX (X= 1,2,3 ..n) resistors.
x.26	SONOS Area: DNWell outside HVI regions overlapping coreid layer overlapping tunm SONOS Flash Macros Array: A continuous piece of n+ diff with at least 34 poly gates
x.27	“areaid.sigPadDiff” encloses any diffusion connected to signal pad through < 10K Ohm resistor.
x.28	areaid.sigPadWell encloses any well connected to signal pad. areaid.sigPadMetNtr encloses any diffusion metallically connected to signal pad. diffusion connected metallically to a signal pad, need both layers (areaid.sigPadDiff, areaid.sigPadMetNtr) over those diffusions. diffusion connected to a signal pad via a resistor, need only areaid.SigPadDiff over those diffusions Signal pad diffusion is any diff or tap connected to a signal pad either metallically or through a resistor of less than 10KOhm. Notes: 1) Only resistors labeled with text.dg as ‘10KOhm’, or connected to pwr/gnd supplies, will break the connectivity between pad and diffusion. 2) Drain Extended FETs must be checked manually for all guard ring rules 3) Legacy IP with ‘100KOhm’ text.dg label will also break the connectivity between pad and diffusion.

CAD switch	Description
local_sub	Creates an isolated substrate in the places where areaid.substrateCut is drawn.
block_level	Promote block pins to be treated like pad connections. Switch should be used at the level of IP Blocks that contain pins which get connected to pad at a higher level. NOTE: Use of the block-level switch checks the following rules: Section 4, Rules 7 and 8 and Section 6, Rules 4.1 and 4.1a. All other rules are not affected.
OLD_TAPPING_RULE_RevU	Checks tapping rules to rev *U of the MTDR, which was based on spacing to signal pad connected diff. To be used ONLY on legacy IP blocks to avoid IP rework.
ACTIVE_ESD	Used with active ESD methodology. Sections 6 through 10 are not checked (Snapback ESD approach). Sections 1 through 5 (Latch-up RULES) and Section 11 (Misc rules) are checked along with the S8 ACTIVE ESD DESIGN RULES (ESD MTDR) SPEC (001-54757)

CAD LuRes switch	Description
block_level	Promote block pins to be treated like pad connections. Switch should be used for IP Blocks that contain pins which get connected to pad at a higher level. NOTE: Use of the block_level switch affects the following rules: Section 4 Rule 5 and Rule 15. At the IP block level this is used for information purposes only.

text.dg Label	Description
probe-only	Identifies bond pads used only for un-bonded pads (characterization/probe) without ESD protection (Section 1, rule 5 and Section 6, rule 1, Section 10, rule 1)
RF	Used to identify RF pads and associated ESD protection scheme (Section 6, rule 6)
10KOhm	Identifies minimum of 10K ohm resistance between signal pad and diffusion. Used to break signal pad to diff connectivity (Section 4, rule 1)
250Ohm	Identifies deep nwells resistively isolated from bond pads by a minimum of 250 ohms. Allows exemption from grounded DNW rule (Section 4, rule 2.1b)
1250ohm	Identifies input paths isolated from bond pads by a minimum of 1250 ohms and series transmission gate. Allows exemption from local ESD protection rule (Section 7, rule 2)
leaker	Used to identify a resistive pull-down leaker arrangement between io net and power or ground nets.
pwr_bias & gnd_bias	Used to short resistor dividers for certain rule checks. Text must be placed over appropriate resistors.
openRes	During latchup checks, resistors short the two nets for their true connectivity. Resistors texted as openRes in text.dg will open the resistor terminals.
switched_power	Identifies the switched power output of a p-channel switch. Label needs to be placed over the metal1 that connects the drain of the p-channel switch.
switch_to_global_bus	Identifies the gates of pass transistors that connect a pad to a common global bus. Use this label ONLY when several pads on a chip connect to the common global bus causing short circuits between the pads.

Section 1: Latch-up Generic Rules		Value	Design Signature Initials	Date
1	DELETED			

Note1: All IP blocks will be checked to the 6 µm tapping rule (except rule 2.3) regardless of spacing to signal pad diff. If the 6 µm tapping rule cannot be met for certain diffusions in the IP block due to concerns of die area increase, then areaid.ltd can be drawn over that portion of diffusion that violates the 6 µm tapping rule. Usage of areaid.ltd triggers the 15 µm tapping rule checks in that region during verification at IP block level.

Usage of areaid.ly on an IP block helps chip-level integrators in the floor-planning stage to assist in placement of IP blocks. Designers should use the areaid.ltd with discretion and not blanket the whole IP block to overcome the 6 µm tapping rule. While strongly discouraged, frozen/legacy IP blocks can default to the original, proximity-based tapping checks by using the “OLD_TAPPING_RULE_RevU” switch.

Note2: CADflow algorithm treats nwell in isolated PWELL or PSUB as a barrier. Distance check goes around the nwell.
Note3: This check is for diffusion metallically connected to signal pad only.

		SPACE < 50 µm from signal pad diff*	SPACE ≥ 50 µm from signal pad diff*
2	Max spacing from center of ptap licon to any part of ndiff within the same psub or pwell	6 µm	15 µm
2.1	Max spacing from center of ptap licon to any part of ndiff within the same isolated pwell (in deep nwell) where the deep nwell, or any nwell that abuts it, does NOT contain a positive power supply connected pdiff	15 µm	15 µm
Exemptions to (2): ESD diffusion; memory array			
3	Max spacing from center of n+ tap licon to any part of p+ diff within the same Nwell or DNW	6 µm	15 µm
Exemptions to (3): ESD diffusion; memory array			

		Value	Initials	Date
4	Minimum distance from diffusion metallically connected to a signal pad to the closest cell in a memory core.	50 µm	**DRC**	

5	Minimum space between ndiff to unrelated ndiff within a common psub or common pwell pdiff to unrelated pdiff within a common nwell or common deep nwell metallically connected to separate pads or external nets (ground, power or signal) Exemptions: 1) FETs covered by areaid:ed 2) FETs in the s8_esd library 3) Diffs connected to probe-Only pads identified by “probe-only” in text.dg layer 4) Source and drain diffs associated with a single FET element (connected across poly gates), whether that FET element is built in a single or multiple active regions 5) s8usbpd_300msw_nsw 6) s8fpiom0s8_top_lvc_b2b_wopad 7) s8fpiom0s8_top_hvclamp_wopad	3 µm	**DRC**
6	Deleted		
7	Minimum distance from diffusion metallically connected to a signal pad to sensitive circuitry (marked by areaid.inj). ONLY the circuitry deemed sensitive should be enclosed by this layer. Note: DELETED	100 µm	**NDRC**
8	Deleted		
9	Deleted – Rule covered by Section 6 Rule 1		

Section 2: SONOS-Specific Latch-up Rules

	Value	Design Signature Initials	Date
Section Deleted – moved to ESD Best Practices Spec			

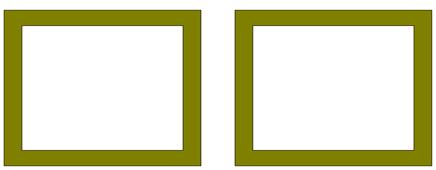
Section 3: Super High Voltage Latch-up Rules

Section deleted

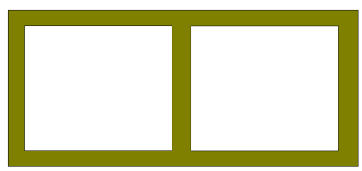
Section 4: Signal Pad Latch-up Rules

		Design Signature		
		Value	Initials	Date
1	Moved to definitions and terminology section			
1.1	These are the allowed guard ring scenarios;	DRC		

	Signal pad connectivity			Inner GR			Outer GR		
	Pad 1	Pad 1 or 2	Common Well/Substrate	Separate	Shared	Common	Separate	Shared	Common
A	Pdiff or Ptap	Pdiff or Ptap	NO	YES	NA	NA	YES	YES	NO
B	Pdiff or Ptap	Pdiff or Ptap	YES	YES	YES	YES	NA	NA	YES
C	Pdiff or Ptap	Ndiff or Ntap	NO	YES	NA	NA	YES	NA	NA
D	Pdiff or Ptap	Ndiff or Ntap	YES	NA	NA	NA	NA	NA	NA
E	Ndiff or Ntap	Pdiff or Ptap	NO	YES	NA	NA	YES	NA	NA
F	Ndiff or Ntap	Pdiff or Ptap	YES	NA	NA	NA	NA	NA	NA
G	Ndiff or Ntap	Ndiff or Ntap	NO	YES	NA	NA	YES	YES	NO
H	Ndiff or Ntap	Ndiff or Ntap	YES	YES	YES	YES	YES	YES	YES



Separate Guard Rings



Shared Guard Rings

Common Guard Ring

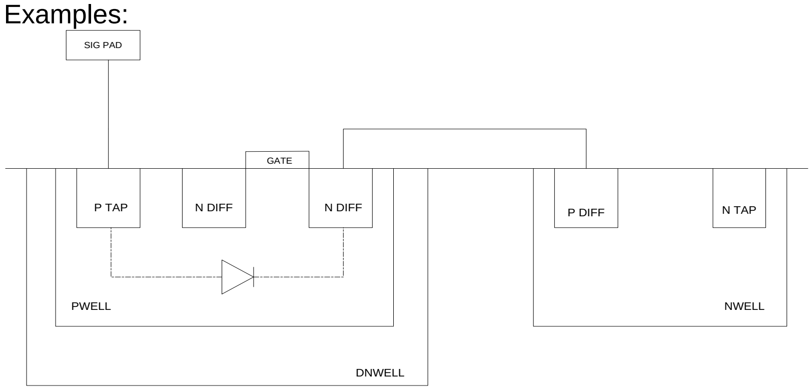
Exemptions:

1. Pad-Pwr ESD Diodes (Appendix A Table 11)
2. Drain Extended Fets (devices covered by areaid.en)
3. s8esdg4_net_d1_420_aup
4. s8esdg4_net_d2_360_sub_aup
5. s8esdg4_net_d4_60_aup

	Section 4: Signal Pad Latch-up Rules	**Value**	**Design Signature** **Initials**	**Date**
2	All signal pad connected n+ diffusion or n+ tap in Nwell must be separated from internal circuitry and any p+ diffusion by a pair of guard rings (See Figure 1). Exemptions: 1. s8esdg4_net_d1_420_aup 2. s8esdg4_net_d2_360_sub_aup 3. s8esdg4_net_d4_60_aup 4. s8blerf_top		**DRC**	
2.1	(A) For signal pad connected ndiff in psub or ntap in nwell, inner guard ring is p+ tap in substrate connected metallically to a ground supply and outer guard ring is n+ tap in nwell connected metallically to a positive external power supply. (B) For signal pad connected ndiff in pwell, inner guard ring is p+ tap in pwell connected metallically to a ground supply or vnb and outer guard ring is n+ tap in dnwell connected metallically to a positive external power supply. Note: DELETED Exemptions to (B) (1) If the ndiff and pwell that contains the ndiff are shorted together, the inner guard ring is p+ tap in pwell connected to signal pad. The outer guard ring is n+ tap in dnwell connected metallically to a positive external power supply. (2) If the signal pad connected ndiff in pwell is after an approved ESD resistor, the inner guard ring is p+ tap in pwell connected to a ground supply, vnb, or the source or drain of the NFET. The outer guard ring is n+ tap in dnwell connected metallically to a positive external power supply. (3) If the pwell that contains the pad connected ndiff is not connected to Vss, and there are no grounded ndiff in the pwell, then the inner guard ring is ptap connected to an internal signal. An internal signal is a signal that is not metallically connected to a pad, it is the source or drain of another FET. (4) indus_top* (5) rainier_top (6) quadspinvsram_top		**DRC**	
2.1a	Signal pad connected deep nwell is not allowed.		**DRC**	
2.1b	Deep nwell tied to ground through a resistance of <250 Ω is not allowed. Only resistors labeled with text.dg as ‘250Ohm’ will break the connectivity between ground and deep nwell		**DRC**	
2.2	deleted - removed all references to super high voltage			

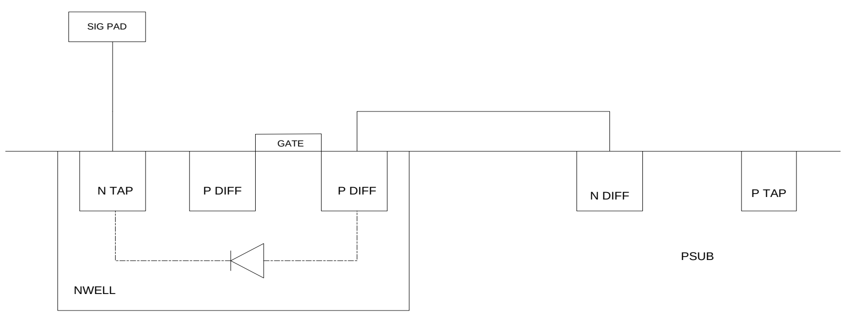
	Section 4: Signal Pad Latch-up Rules	Value	Design Signature Initials	Date
3	All p+ signal pad connected diffusion or ptap in pwell (or p+ tap in DNWell as part of a DE_PMOS device) must be separated from internal circuitry and any n+ diffusion by a pair of guard rings (Figure 1). Exemptions: 1. s8esdg4_net_d1_420_aup 2. s8esdg4_net_d2_360_sub_aup 3. s8esdg4_net_d4_60_aup		DRC	
3.1	For signal pad connected p+ diffusion, or ptap in pwell, inner guardring is n+ tap in Nwell connected metallically to a positive power supply and outer guardring is p+ tap metallically connected to external ground supply (Figure 1). Exemptions: 1) pdiff and the nwell in which that pdiff resides are connected to the same signal pad 2) s8_esd_localdiode_lv, s8_esd_localdiode_hv 3) Non_Vcc_nwell		DRC	

Section 4: Signal Pad Latch-up Rules	Value	Design Signature Initials	Date
3.2 Any diffusion (P+ or N+) connected to signal pad via a parasitic diode must be double guard ringed. Examples:		______	______



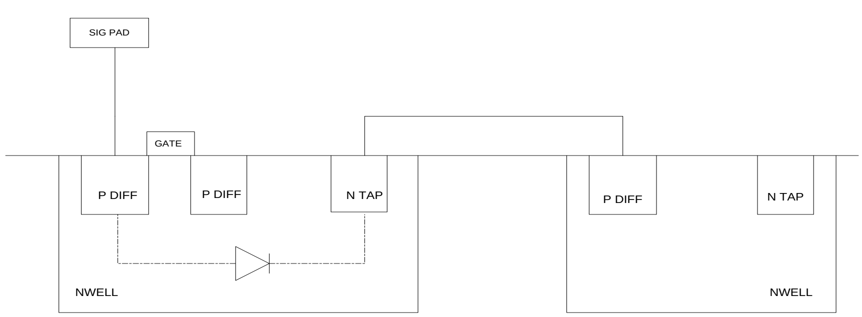
Signal Pad connected Pwell
(requires double guard ring per)

Parasitic diode connected Pdiff
(requires double guard ring)



Signal Pad connected Nwell
(requires double guard ring per)

Parasitic diode connected Ndiff
(requires double guard ring)



Signal Pad connected Nwell
(requires double guard ring per)

Parasitic diode connected Pdiff
(requires double guard ring)

	Section 4: Signal Pad Latch-up Rules	Value	Design Signature Initials	Date
4	Each guardring must be continuously strapped in li. Exemptions: 1) s8ppscio_sio_pudrvr_reg_pu 2) p3ag_gpio_amx 3) p3ag_gpio_ag 4) s8psoc3io_sio_pudrvr_reg_pu Additionally, the guardring must be strapped in Met. The Met may be broken, but the total series resistance from any part of guardring w/o met to the portion of met strap connected to supply bus must be less than 125 Ω (connected through li). Guardrings located on the interior/circuit side of the ESD resistor also require metal strapping,		DRC ______	______

	Section 4: Signal Pad Latch-up Rules	Value	Design Signature Initials	Date
5	The outer guard ring must connect to respective external power or ground pad Maximum metal resistance from outer guard ring to respective power or ground pad. a) Around diffusion connected to signal pad in metal b) Around diffusion connected to signal pad via ESD resistor Exemptions: 1) psoc4able256_top 2) psoc4al_top 3) psoc4ads2_top 4) psoc4a_top 5) psoc4able_top 6) fpg1_top 7) tsg5_m_top c) Within 50 µm of any signal pad connected diffusion, maximum resistance of any outer guard ring hook-up to its respective supply pad Exemptions: 1) s8ctbm_opa 2) psoc4able256_top 3) psoc4able_top 4) s8atlasana_top 5) s8tsafg6*_top 6) tsg6m_top 7) m0s8gen4_top 8) fpg1_top 9) psoc4a_top 10) psoc4ads2_top 11) psoc4al_top 12) ccg4_top 13) tsg5_m_top Use NOT_GR text.dg on inner guard ring if the device has more than 2 guard rings Note: This is a chip level check; however the block_level switch may be used to run this on an IP block for information only.	 5 Ω 10 Ω 10 Ω	**LuRes**	
5.1	deleted - removed all references to super high voltage			
6	Minimum width of strapping for outer guardring around diffusion that is metallically connected to signal pad. Li M1	 0.85 µm 0.65 µm	**DRC**	

	Section 4: Signal Pad Latch-up Rules	Value	Design Signature Initials	Date
6.1	Minimum width of strapping for inner guardring around diffusion that is metallically connected to signal pad. Li M1 Exemptions: 1) s8_esd_signal_40_sym_hv_2k_dnwl_aup1_b	 0.85 µm 0.65 µm	**DRC**	
7	Minimum width of metal hooking outer guardring to power or ground bus for those elements that are metallically connected to signal pad.	3 µm	**DRC**	
7.1	deleted - removed all references to super high voltage			

8	Minimum number of vias connecting any two metal layers from outer guardring to power or ground bus for those elements that are metallically connected to signal pad.	6	**DRC**
	Max space between licons for strapping/width checks in rules 6,7, and 8	2 µm	**DRC**
	Max space between mcons for strapping/width checks in rules 6,7, and 8	2 µm	**DRC**
	Max space between vias for strapping/width checks in rules 6,7, and 8	2 µm	**DRC**
	Max space between vias2 for strapping/width checks in rules 6,7, and 8	2 µm	**DRC**
9	Minimum width of outer guardring taps for those elements that are metallically connected to signal pad.	0.85 µm	**DRC**
10	Deleted		
11	Deleted		
12	For all flows - Minimum spacing between:	27 µm	
	a) pdiff, metallically connected to signal pad, and grounded ndiff Exemptions: 1) The following cells: s8sio_lvttl2k_io_r, s8sio_gp4k_outbuf 2) The pdiff and the nwell which it resides in are connected to the same signal pad		
	b) pwell, metallically connected to signal pad, and grounded ndiff	40 µm	
	c) pdiff, metallically connected to signal pad, and grounded nwell Exemptions: 1) The pdiff and the nwell which it resides in are connected to the same signal pad	40 µm	
	d) pwell, metallically connected to signal pad, and grounded nwell	40 µm	
	e) ndiff, metallically connected to signal pad, and positive power supply connected pdiff Exemptions: 1) The ndiff and the pwell which it resides in are connected to the same signal pad	27 µm	**DRC**
	f) nwell, metallically connected to signal pad, and positive power supply connected pdiff	40 µm	
	g) ndiff, metallically connected to signal pad, and positive power supply connected pwell Exemptions: 1) The ndiff and the pwell which it resides in are connected to the same signal pad	40 µm	
	h) nwell, metallically connected to signal pad, and positive power supply connected pwell	40 µm	
	i) pdiff, metallically connected to signal pad, and ndiff, metallically connected to a different signal pad	27 µm	
	j) pdiff, metallically connected to signal pad, and nwell metallically connected to a different signal pad Exemptions: 1) The pdiff and the nwell which it resides in are connected to the same signal pad	40 µm	
	k) pwell, metallically connected to signal pad, and ndiff metallically	40 µm	

	connected to a different signal pad	
l)	pwell, metallically connected to signal pad, and nwell metallically connected to a different signal pad	40 µm
m)	ndiff, metallically connected to signal pad, and pdiff in an *At RISK Non_Vcc_nwell* (x.14a) Exemptions: 1) The ndiff and pdiff are metallically or resistively connected to the same net – default to the 27 µm rule 2) The *At RISK Non_Vcc_nwell* contains pdiff metallically or resistively connected to the signal pad connected ndiff 3) The ndiff and the pwell which it resides in are connected to the same signal pad 4) The *At RISK Non_Vcc_nwell* contains pdiff metallically or resistively connected to the *At RISK Non_Vcc_nwell*	33 µm
n)	ndiff, metallically connected to signal pad, and *At RISK Non_Vcc_nwell* (x.14a) Exemptions: 1) The *At RISK Non_Vcc_nwell* contains pdiff locally shorted to the *At RISK Non_Vcc_nwell* on die in metal through ≤ 1 Ω 2) The *At RISK Non_Vcc_nwell* contains pdiff metallically or resistively connected to the signal pad connected ndiff 3) The ndiff and the pwell which it resides in are connected to the same signal pad 4) The *At RISK Non_Vcc_nwell* contains pdiff metallically or resistively connected to the *At RISK Non_Vcc_nwell* Note: Each pdiff must be checked independently. If any pdiff fails to meet the criteria in 1) or 2) then this spacing rule must be met	16.75 µm
o)	pdiff, metallically connected to signal pad, and nwell connected to 1.8V (LV) or lower Exemptions: 1) atlas_top 2) fpg1_top 3) ccg2_top 4) ccg4_top 5) s8sarmux_top 6) s8usbfsm0s8_top 7) psoc4a_top 8) psoc4al_top 9) psoc4ads2_top 10) psoc4able_top 11) psoc4able256_top	27 µm
p)	pwell, metallically connected to signal pad, and nwell connected to 1.8V (LV) or lower Exemptions: 1) s8fpafeg1_top 2) s8fpafeg1_io_rx_2x1 3) s8tnvra_psoc3_osc32a_p3 4) m0s8gen4_top	40 µm

Rule 12 Notes:

1) If nwell can be externally grounded (i.e. customer driven) then grounded

	nwell rules must be followed.	
12.1	deleted - removed all references to super high voltage/drain extended devices	
13	Outer n-type guardrings around signal pad n-type injectors must not contain any PMOS device. i.e. PMOS is not allowed between the inner tap ring and the outer tap ring; PMOS is not allowed in the shared well of the outer ntap ring Exemptions: 1) s8tnvra_psoc3_osc32a_p3 2) p3ag_asws 3) s8p5ana_upper_region_nopumps 4) s8tnvra_psoc3_osc32a_p3	**DRC**
14	Diffusion resistors are not allowed within guardrings around signal pad diffusions. Exception is when resistance of a source/drain is modeled as a diffusion resistor (i.e. diff.res is drawn over srcdrn region). i.e. Diff resistor is not allowed between the inner tap ring and the outer tap ring; Diff resistor is not allowed in the shared well of the outer ntap ring.	**DRC**
15	Any circuitry within 100 µm of any diff or tap metallically connected to a signal pad must have its source and body shorted locally. (Shorted locally implies source/body connected at the FET in metal. No separate routes for source and body) LatchupRes measures the resistance between source/body of FETs identified in this rule. The LatchupRes results must be reviewed by the Design team and any anomalous high resistance values **(above 100 Ω)** corrected via layout. Initial and date here indicating completion of this activity. The final results must be available and presented at the MTDR Layout Review. Note: This is a chip level check; however the block_level switch may be used to run this on an IP block for information only.	**LuRes** _____ _____
16	Deleted	

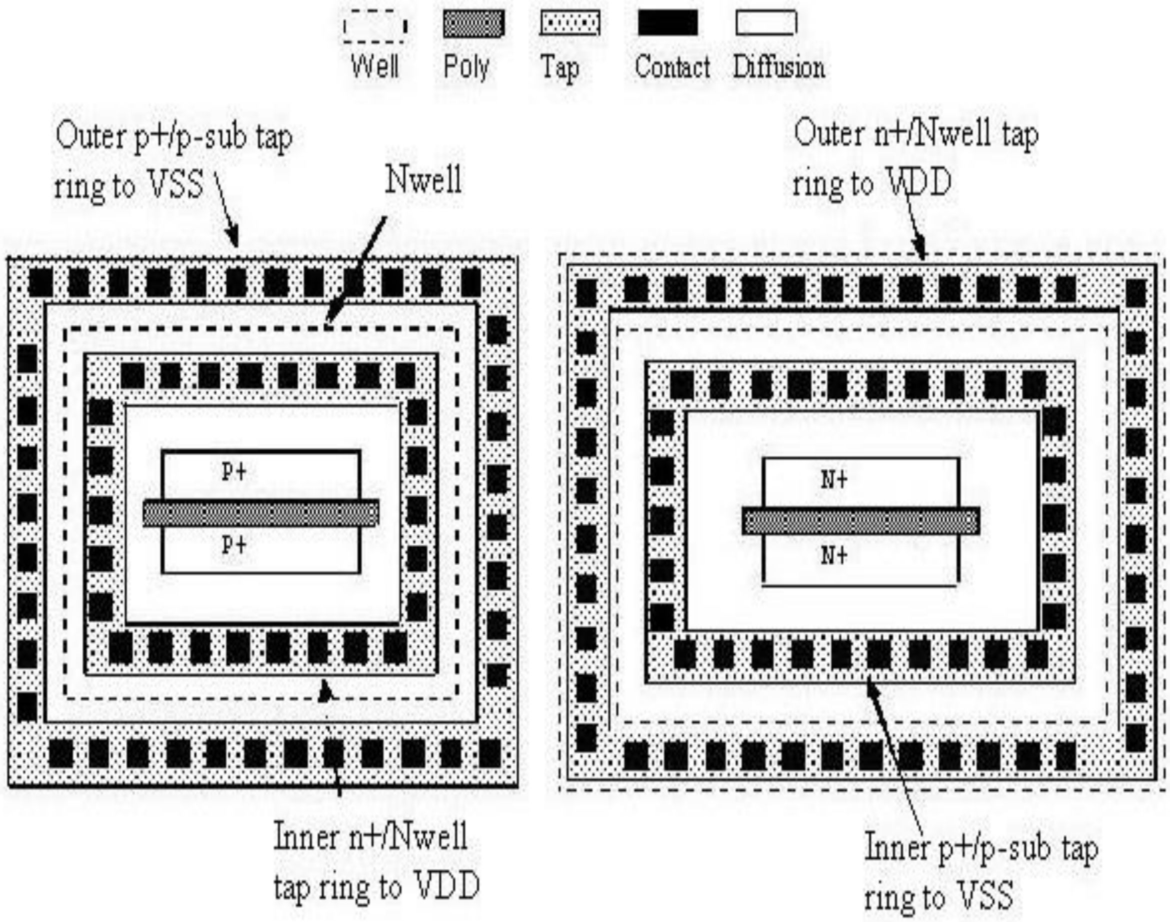


Figure 1 – Signal Pad Connected Diff Guard-ring Construction

	Section 5: Latch-up Special Cases	**Value**	**Design Signature Initials**	**Date**
1	*At RISK Non_Vcc_nwell* (x.14a) must adhere to the rules in this section. Exemptions: 1) If the pdiff and the Non_Vcc_nwell that contains it are locally shorted on die in metal through ≤ 1 Ω 2) If the *At RISK Non_Vcc_nwell* is > 50µm from a signal pad metallically connected ndiff or nwell 3) Cells: s8bbcnv_psoc5_top_18 p3ag_p_sio Any updates to this IP must be reviewed with the ESD/LU group Notes: 1) For exemption, CAD flow checks for source/body locally shorted in metal. It is the Design Team's responsibility to validate resistance measurement. 2) A flat file can be obtained from the CAD group to help identify Non_Vcc_nwells	**DEF**		
1a	This Nwell must be guardringed by a p+ tap inner ring connected to Vss and continuously strapped in li. Additionally, the guardring must be strapped in Met. The Met may be broken, but the total series resistance from any part of guardring w/o met to the portion of met strap connected to supply bus must be less than 125 Ω (connected through li). Note: Guard Ring can be shared with other guard rings.		**DRC** _____	_____
1b	This Nwell must further be guardringed by an n+ tap/Nwell outer ring connected to a positive external power supply and continuously strapped in li. Additionally, the guardring must be strapped in Met. The Met may be broken, but the total series resistance from any part of guardring w/o met to the portion of met strap connected to supply bus must be less than 125 Ω (connected through li). Note: Guard Ring can be shared with other guard rings.		**DRC** _____	_____
1c	Rule deleted			
2	If inherent parasitic PNP bipolar transistor is used, surround the device or blocks of devices by an inner p+ tap guardring tied to Vss. Additionally, this structure must be guardringed by a n+ tap/Nwell outer guardring connected to a positive power supply. Both rings must be continuously strapped in li. Additionally, the guardrings must be strapped in Met. The Met may be broken, but the total series resistance from any part of guardring w/o met to the portion of met strap connected to supply bus must be less than 125 Ω (connected through li).		**DRC** _____	_____
	Max space between licons for rules 1a, 1b and 2 **Exemption**: s8p3ana_top	2 µm	**DRC**	

	ESD_At_Risk_Wells Defined as; 1) Nwell that contains pad connected diffusion and the well is not metallically connected (hard tied) to a Vcc net 2) Pwell in DNW that contains pad connected diffusion and the well is not metallically connected (hard tied) to a Vss net		
5	The equivalent resistance from the pad to the diffusion in the well must be ≥ 75 Ω (See Figure 1a) Exemption: Pad connected ESD diffusion – the ESD diffusion must be in a separate/isolated well, it cannot be in the same well as any other resistively isolated diffusions. (See Figure 1b)	______	______
6	Any At Risk Non-Vcc nwell which has p+ diffusion connected to a power supply 3.3V or higher should have its own p+ tap ring connected to ground and there should not be any n+ diffusion between the p+ diffusion in the At Risk Non-Vcc nwell and the p+ tap ring. Exemptions: s8p3iomacro_gpio_macro s8p3iomacro_gpio_macro_ao s8p3iomacro_sio_macro rainier_top		**DRC**
7	Grounded nwell requires a substrate tap ring around it metallically connected to the ground. Exemptions: s8iom0s8_top_lvc_b2b_wopad s8iom0s8_top_lvclamp s8atlasana_esd_gnd2gnd_sub_dnwl		**DRC**

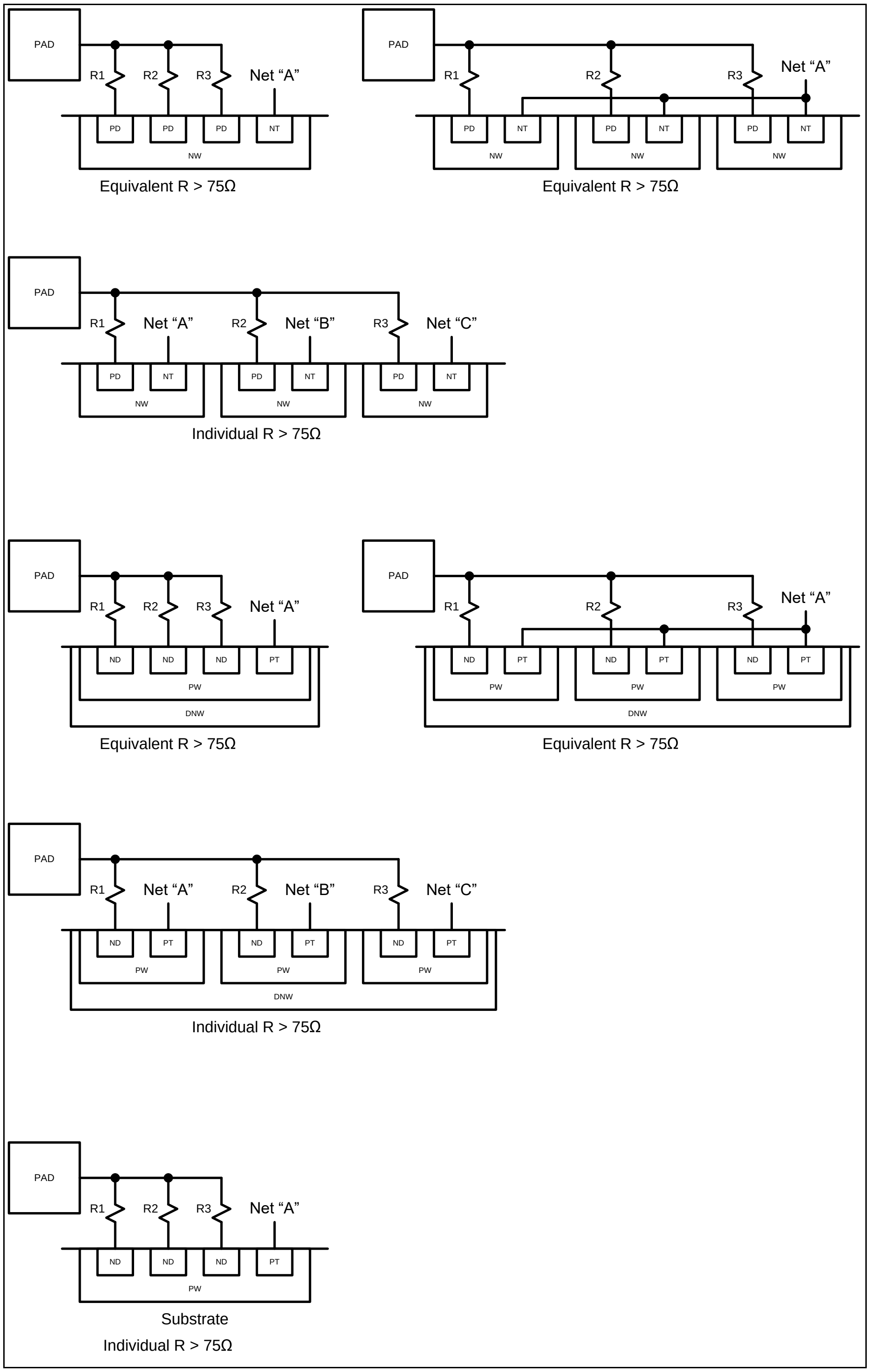


Figure 1a (Rule 5)

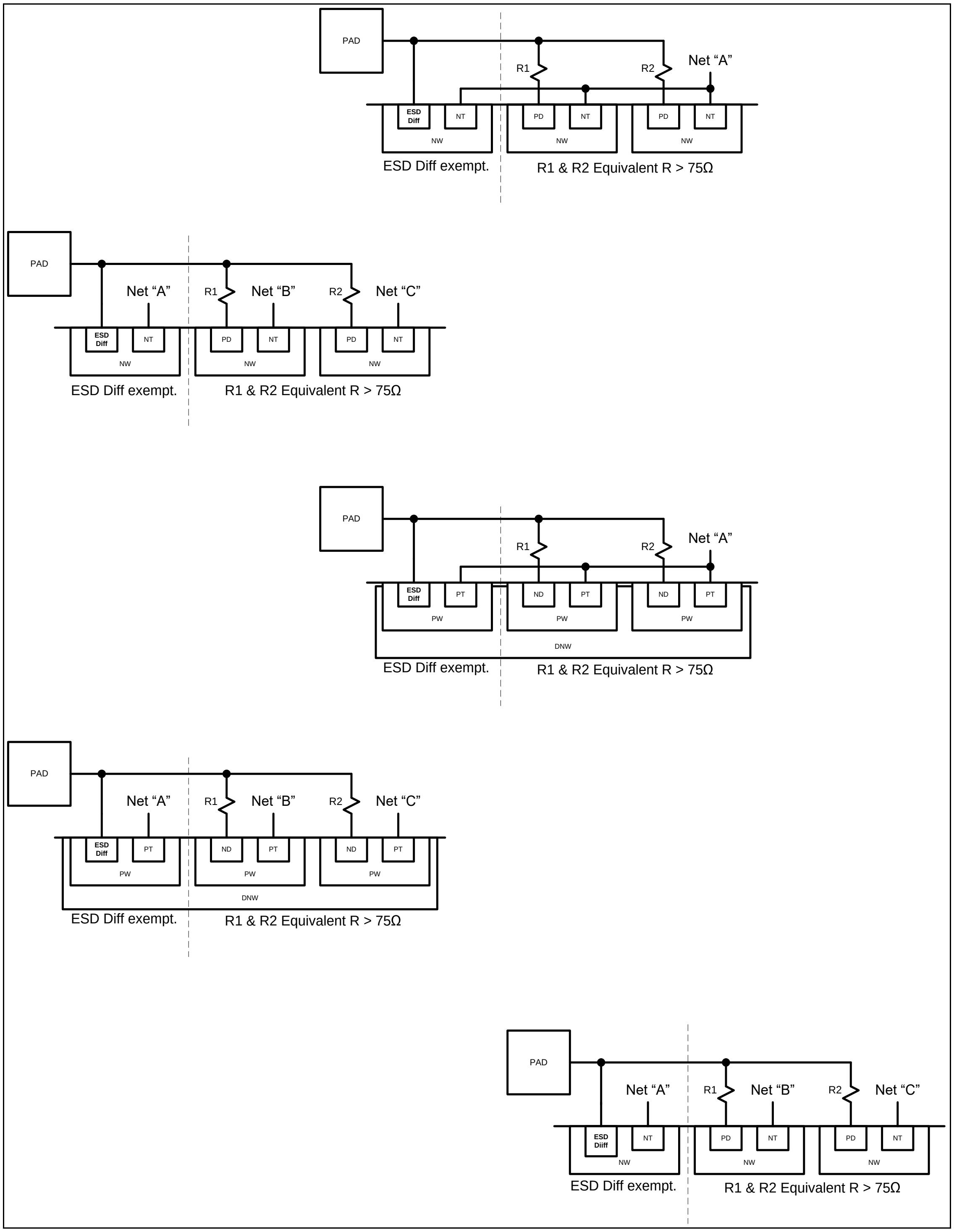


Figure 1b (Rule 5 Exemption)

	Section 6: Signal Pad Main ESD Protection	Value	Design Signature Initials	Date
1	Any signal pad that will be bonded out in any package must have ESD protection devices from the ESD library connected metallically to the pad. Note: Probe-Only pads without ESD protection must have "probe-only" in text.dg layer Notes: 1) VCSEL drivers require additional clamps (a diode and a FET) between VCSEL pins in addition to the standard pad to ground clamps (see Figure 2 and Appendix A, Tables 1, 10, and 11) 2) Designer must specify VCSEL pads in the "VCSEL driver pairs" field of the latch-up DRC form in the format "(VCSEL_out_pad VCSEL_rtn_pad)".		**DRC**	
2	Deleted			
3	**For 5V tolerant input only pads**, the main ESD protection device is a soft-tied gate NFET. For allowable existing structures, see Appendix A. Note: Designer must specify this pad in the "Inp Pad Nets (5v)" field of latchup DRC form.		**DRC**	
3.1	**For SONOS programming pad**, the main ESD protection device is a soft-tied gate NFET. For allowable existing structures, see Appendix A Note: Designer must specify this pad in the "Vpp Pad" field of latch-up DRC form.		**DRC**	
4	**For 5V tolerant output or I/O pads**, the main ESD protection is performed by the output driver(s) For allowable existing structures, see Appendix A. Note: Designer must specify this pad in the "Output or I/O Pad Nets (5v)" field of latch-up DRC form.		**DRC**	
4.1a	Unused (spare) poly gates in the Output or I/O pad output NMOS and PMOS drivers must be soft-tied through a poly resistor to Vss and Vcc respectively.		**DRC**	
4.1b	The minimum value of this resistor is **…**	1000 Ω	_____	_____
5	deleted - removed all references to super high voltage			
6	For RF pad specific info refer to "ESD and LU Best Practices Spec 001-44971"		**RF Pad Specific Information**	

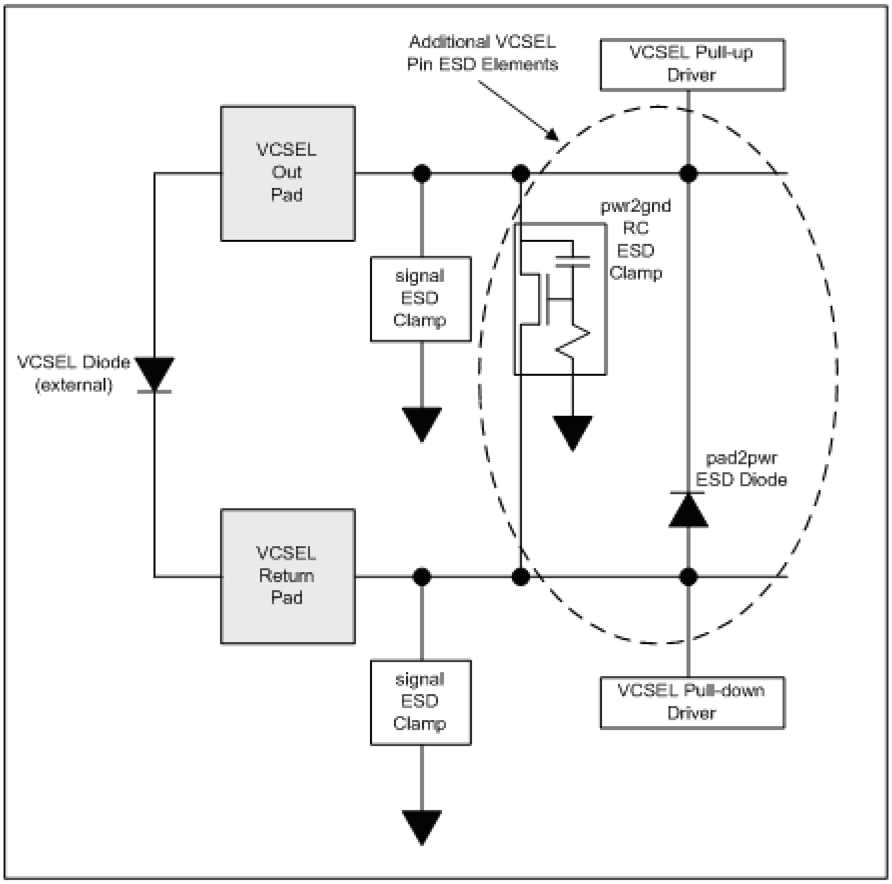


Figure 2 – VCSEL Pad ESD Protection Scheme

	Section 7: Signal Pad Secondary ESD Protection	Value	Design Signature Initials	Date
1	Any diff, tap or gate, not part of a main ESD structure or VCSEL driver described in Section 6, that is connected to a signal pad must be separated from the pad by an ESD poly resistor. (Valid main ESD structures documented in Tables 1, 10, and 11) EXEMPTION: s8linbuf_top s8linbuf_p_top		**DRC**	
1.1	Minimum resistance of ESD resistor if diff, tap or gate in rule 1 is part of 110A FETs only.	74 Ω	_____	_____
1.2	Minimum resistance of ESD resistor if diff, tap or gate in rule 1 is part of 32A FETs only.	248 Ω	_____	_____
1.3	ESD resistor is from ESD library. See Appendix A.		**DRC**	
2	On the circuit side of the input ESD resistor, one of the six approaches in Figure 3 must be used to prevent VIH/VIL shifts caused by ESD stressing. Exemption: HV diff connections are exempted. Exemption: If the total input path resistance is greater than or equal to 1250 Ω (ESD resistor AND additional resistor) **AND** is further isolated by transmission/pass gate(s). The additional resistor must be placed on the circuit side of the ESD resistor. For this exemption to pass the label 1250 Ω must be placed over the ESD resistor using text.dg layer.		_____	_____
a	Approach in figures 3 (a) through (f) (DRC signal to gnd path ONLY)		**DRC**	
b	Deleted			
3	The NFET local clamp is taken from the ESD library. See Appendix A. The following cells are exempted from this rule; s8capio_p1_top s8capio_p1tm_top s8capio_p10tm_top		**DRC**	
3.1	The NFET local clamp connects to the same Vss as that of the input buffer.		**DRC**	
3.2	If input buffer has separate isolated_on_die grounds for its body and source connections, then two separate local clamps are required; one from the input's gate to the input's source and another from the input's gate to the input's body		**DRC**	
3.3	The maximum resistance between the input buffer's NFET/PFET source and the local ESD elements' source/cathode. If resistance (R) is in the primary current path If resistance (R) is NOT in the primary current path (clamp source connected directly at NFET source) **See Figure 4**	1 Ω 5 Ω	______ ______	______ ______
4	The local diode is taken from the ESD library (see Appendix A), and the local diode connects to the same Vcc as that of the input buffer's PFET.		_____	_____
4.1	Deleted – covered in Rule 4 above			
5	For Supervoltage Testmode pad specific info refer to "ESD and LU Best Practices Spec 001-44971"		**Supervoltage Testmode Pad Specific Information**	

6	NMOS transistor poly gates connected to signal pads where the signal pad could be held at a DC level of anywhere between half of Vcc to Vcc inclusive for any time longer than the inverse of the smallest AC signal pad frequency of the product, must have a gate length of at least… Notes: 1) This is not ESD related but is due to hot carrier degradation. 2) Designer must specify in latch-up DRC form (Esd 7_6Nets) which signals should be checked for this rule	1.1x TDR min		**DRC**
7	On pads without a primary diode to VCC (parasitic pfet or primary ESD diode), all secondary LV and HV devices with gates tied to VCC must have a series gate resistor of at least……. This includes HV and LV pass gates, Cascode local clamps and Floating Well PMOS bias devices.	1000 Ω	______	______
8	Minimum equivalent resistance from signal pad to n+ diffusion which is in an isolated pwell where the isolated pwell is not metallically connected to ground or to the same signal pad.	150 Ω	_____	**LuRes** ____

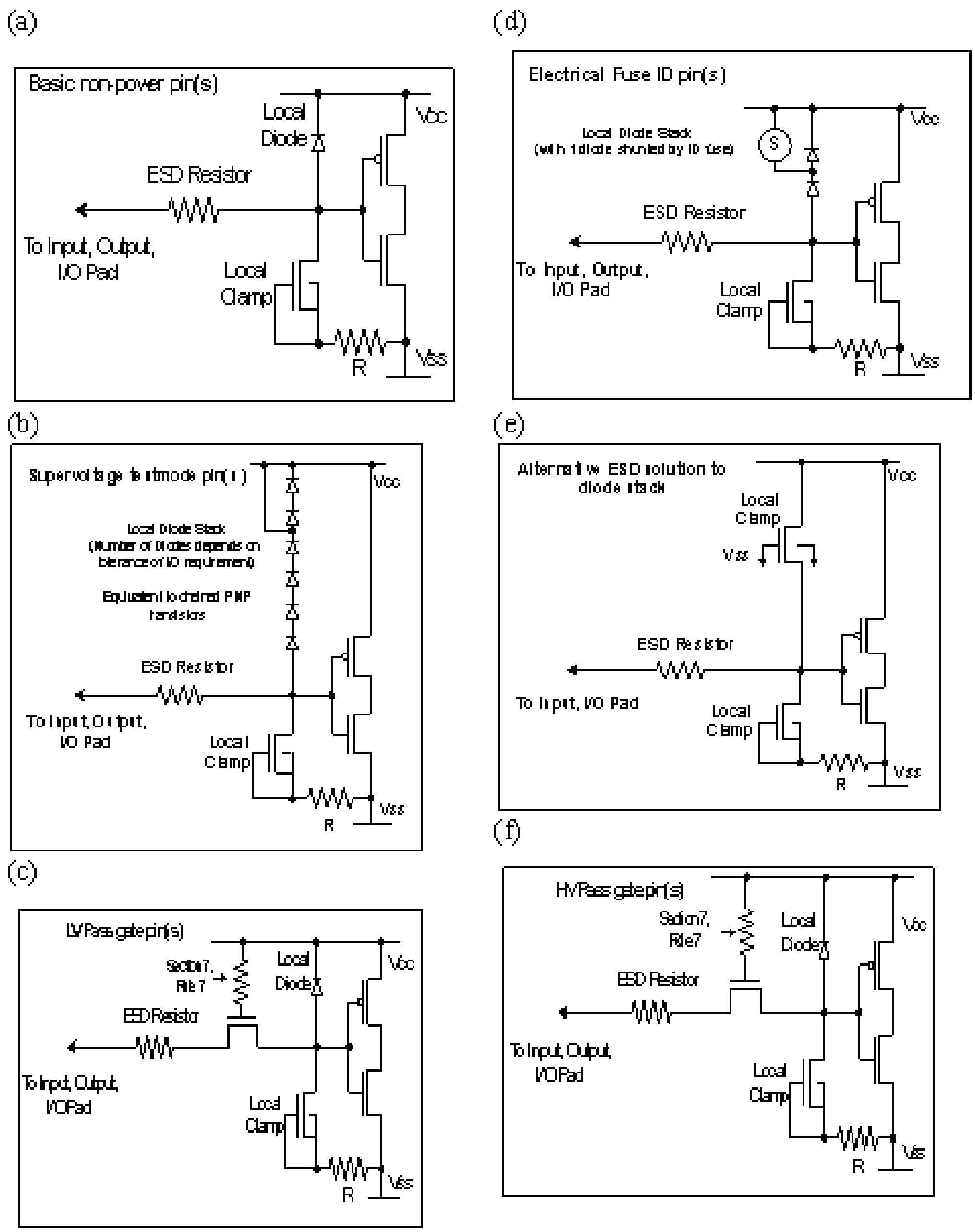


Figure 3 – Secondary Protection Options

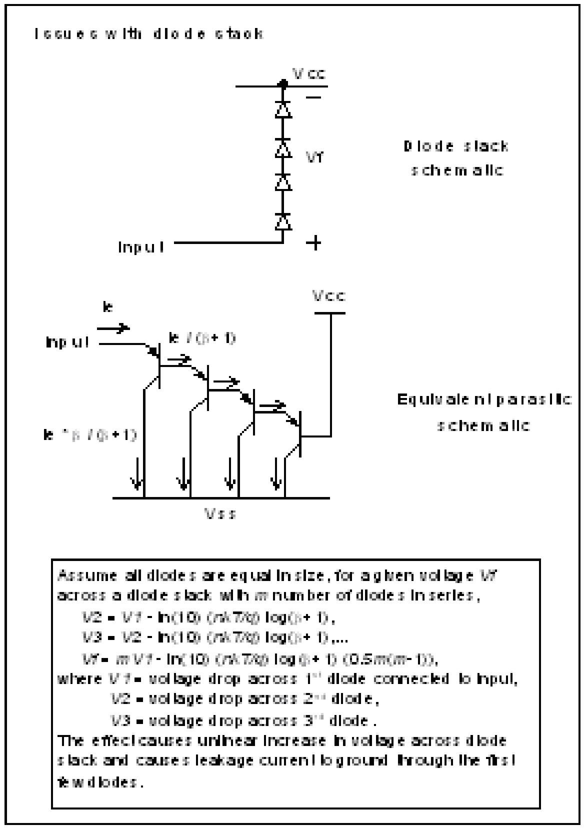
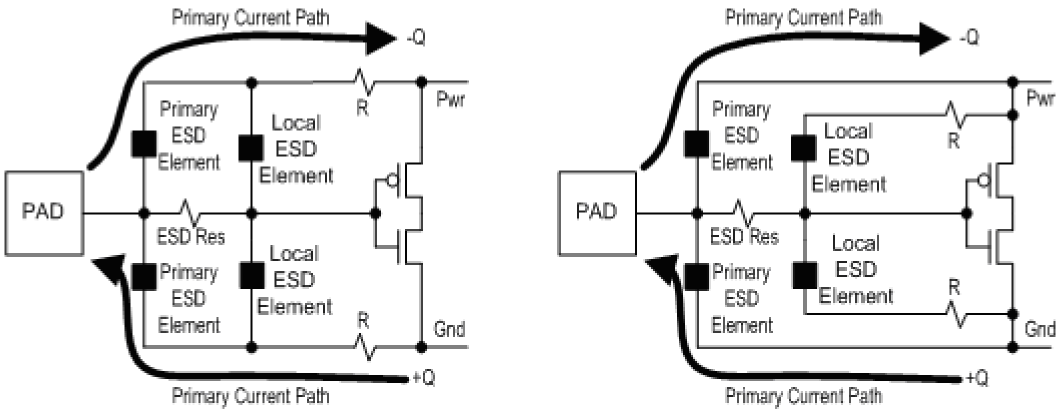


Figure 3 (cont'd) – Secondary Protection Options



Local protection in primary current path

Local protection not in primary current path

Figure 4 – Local Protection (Rule 3.3)

	Section 8: Power Clamping Rules	**Value**	**Design Signature Initials**	**Date**
1	Any ground that is isolated_on_die from the substrate ground (i.e. the one that connects to the substrate through p+ tap and is the highest capacitance ground on the chip) must clamp to the substrate ground. Note: Designer must specify the substrate ground in the “Main Ground Supply” field of the latch-up DRC form.		**DRC**	
1.1	Interground clamp (consisting of bi-directional diodes) is taken from ESD library. See Appendix A. Note: A minimum of one diode in each direction and a maximum of two in one direction with one in the other is allowed. Placement of two series diodes in each direction with the extra one optioned out in metal is recommended.		**DRC**	
1.2	Each independent on-chip ground pad, including substrate pads, must metallically connect to at least one B2B diode. No B2B diode is required if the chip has only one independent external ground. Exemptions: s8tnvsio_vgnd rainier_top indus_top* k2_chip manas_top		**LuRes**	
2	Any power supply that is isolated_on_die from the main externally supplied power supply that has the highest capacitance on chip must clamp to this highest capacitance power supply. Note: Designer must specify the main external power supply in the “Main Power Supply” field of the latch-up DRC form. Exemption: quadspinvsram_top		**DRC**	
2.1	Interpower clamp is taken from ESD library. See Appendix A. - This clamp is a soft-tied grounded gate NFET with source and drain diffs connected to the powers - It has a symmetric layout for drain and source licon to poly space - The 1.8V version has Nwell under both source and drain licons - The gate tie to ground is at least 1000 Ω of poly resistance Note: Designer must specify the Low voltage and high voltage power supply nets in the “Lv Power Supply (1.8v)” and “Hv Power Supply (3.3 or 5v)” fields of the latch-up DRC form Exemption: quadspinvsram_top		**DRC**	
2.2	If both powers are 1.8V, then the low voltage (32A) NFET clamp must be used (For allowable existing structures, see Appendix A.) Exemption: quadspinvsram_top		**DRC**	

	Section 8: Power Clamping Rules	Value	Design Signature Initials	Date
2.3	If one of the powers is 3.3V or 5V, then the high voltage (110A) NFET clamp must be used. Exemption: quadspinvsram_top		**DRC**	
3	Any power supply must clamp to its associated ground. Notes: 1)This also applies to power to ground clamp(s) required for the current steering diode RF pad protection scheme 2) Designer should specify each power supply and its associated ground in “Power Ground Pairs” field of the latch-up DRC form		**DRC**	
3.1	Power to ground clamp is taken from ESD library. See Appendix A. For 1.8V, 3.3V or 5V power supplies: - This clamp is an RC-triggered NFET with source diff connected to ground, drain diff connected to power and substrate connected to the main substrate ground (or to the source ground if source ground to substrate ground connection through the substrate is not a concern) - It has an asymmetric layout for drain and source licon to poly space - The 1.8V version has Nwell under the drain licons only		**DRC**	
3.2	deleted - removed all references to super high voltage			
4	An interpower clamp is required, in addition to rule 2 above, when the following condition occurs: A signal line emanates from a first circuit block powered by one supply and ends in a second circuit block powered by another supply where these two power supplies are isolated_on_die from each other. An interpower clamp is also required between the main externally supplied power supply and the internal regulated power supply that is regulated from the former supply. Note: Designer must specify the power supply pairs that need to have the above conditions checked, in the “Esd 8_4Nets” field of the latch-up DRC form.		**DRC**	
5	Deleted			

Section 9: Bus Resistance Rules

	Note: These rules are checked at the chip level, they are not applicable at the IP block level. However the block_level switch may be used to run this on an IP block for information only.	**Value**	
1	Maximum ground bus resistance between **HV** signal pad's pad2gnd ESD clamp and its associated ground pad.	5 Ω	**LuRes**
2	Maximum ground bus resistance between **LV** signal pad's pad2gnd ESD clamp and its associated ground pad.	2 Ω	**LuRes**
3	Maximum ground bus resistance between any ground pad and its nearest associated pwr2gnd ESD clamp.	5 Ω	**LuRes**
4	Maximum ground bus resistance between any ground pad and its nearest associated gnd2gnd ESD clamp down to Met1. Allowable B2B diodes: s8_esd_gnd2gnd_120x2_lv_isosub s8_esd_gnd2gnd_120x2_lv (Not Recommended) Exemptions: s8bbcnv_psoc3p_top_18 s8tkm0s8_psp_vssa s8tsg4io_top_vssio_2	2 Ω	**LuRes**
5	Maximum power bus resistance between any power pad and its nearest power to power clamp.	5 Ω	**LuRes**
6	Maximum power bus resistance between any power pad and its nearest power to ground clamp.	5 Ω	**LuRes**
*** **7**	Maximum metal resistance from one end of the bus (power, ground) to another	5 Ω	____ ____

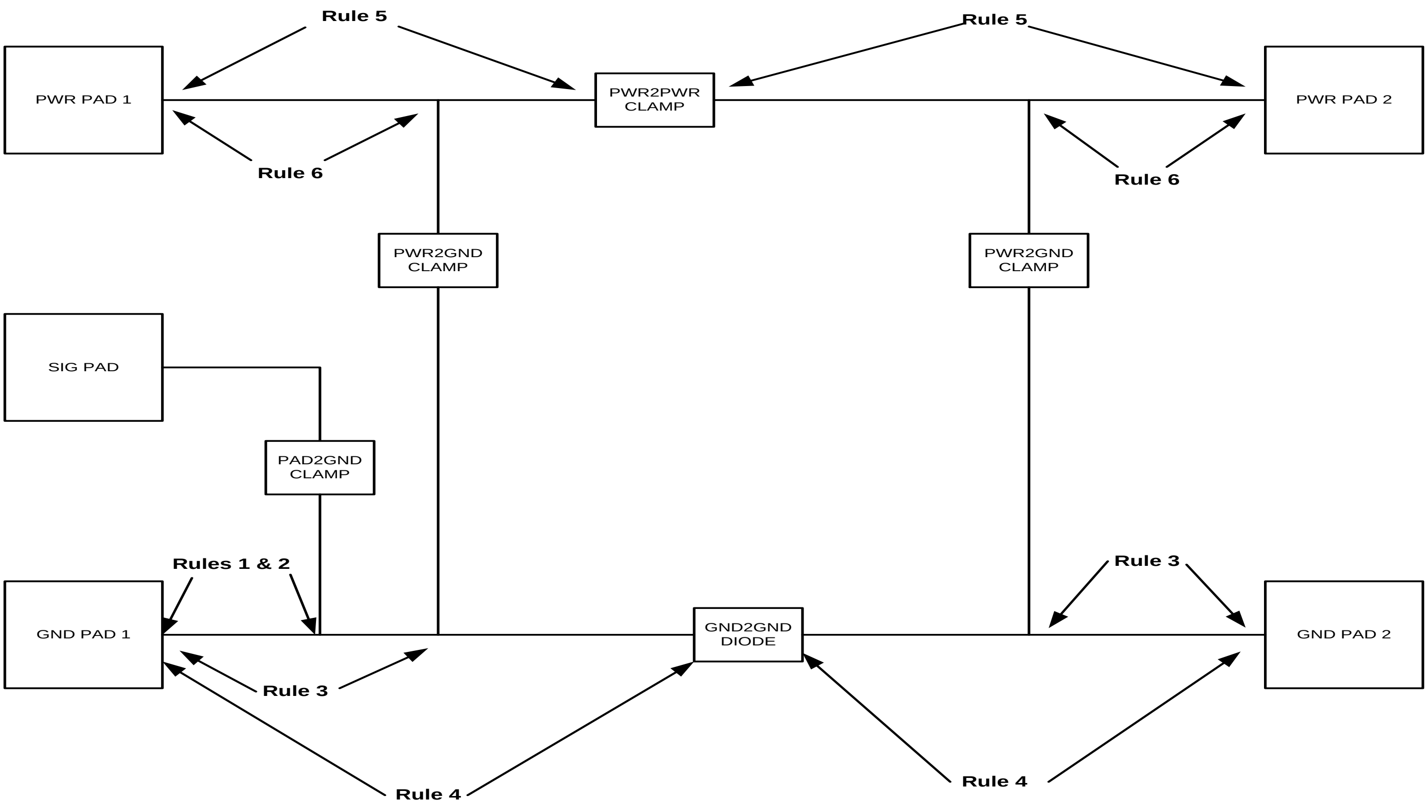


Figure 5 – Bus Resistance Rules

Section 10: Main ESD Clamp Hookup	Value	Design Signature Initials	Date

1 **Any pad/regulator output must be connected to a main ESD protection clamp.**
Min metal width from ESD protection clamp to pad/regulator output. (For 2kV requirements)

Flow	M1	M2	M3	M4	M5
s8d*	12 µm	12 µm	TDR min (INDM)	NA	NA
s8t*	12 µm	12 µm	5 µm	NA	NA
s8q*	12 µm	12 µm	6 µm	2.5 µm	NA
s8p*	12 µm	12 µm	6 µm	6 µm	2.5 µm

Min metal width from ESD protection clamp to pad/regulator output. (For 4kV requirements)

Flow	M1	M2	M3	M4	M5
s8d*	20 µm	20 µm	TDR min (INDM)	NA	NA
s8t*	20 µm	20 µm	8 µm	NA	NA
s8q*	20 µm	20 µm	9 µm	3.5 µm	NA
s8p*	20 µm	20 µm	9 µm	9 µm	3.5 µm

Notes:
1) Min metal width rules apply to the source/drain/body connections of the FET.
2) Probe-Only pads without ESD protection must have “probe-only” in text.dg layer
3) Main ESD clamps include power to power NFETs, power to ground NFETs and interground ESD diodes

Metal width connecting to ESD clamp fingers must be at least… 1.5 µm

Section 10: Main ESD Clamp Hookup

		Value	Design Signature Initials	Date
4	Minimum number of contacts connecting ESD structures to regulator outputs, power buses, ground buses, power pads, ground pads or non-power pad is given below (in the path of the ESD current).		______	______

For 2kV requirements:

Flow	licon	mcon	via	via2	via3	via4
s8d*	400	250	90	15	NA	NA
s8t*	400	250	90	50	NA	NA
s8q*	400	250	90	70	8	NA
s8p*	400	250	90	70	70	8

For 4kV requirements:

Flow	licon	mcon	via	via2	via3	via4
s8d*	400	250	90	20	NA	NA
s8t*	400	250	90	50	NA	NA
s8q*	400	250	90	70	8	NA
s8p*	400	250	90	70	70	8

Notes:
1) See Section 4 for max spacing between contact values
2) This rule does not apply to guardrings around the main ESD clamps

4.1 Deleted (moved to table in rule 4)
4.2 Deleted (moved to table in rule 4)
4.3 Deleted (moved to table in rule 4)
4.4 Deleted (moved to table in rule 4)
4.5 Deleted (moved to table in rule 4)
5.1 Deleted
5.2 Deleted

	Section 11: Miscellaneous Rules	Value	Design Signature Initials	Date
1	**SoftConn Rules**			
1.1	Nwell must be connected to Ntap at least once.		**DRC**	
1.2	Substrate containing diffusion must be connected to Ptap at least once		**DRC**	
1.3	Isolated Pwell must be connected to Ptap at least once.		**DRC**	
1.4	Ptap net in isolated Pwell does not conflict with substrate majority connection.		**DRC**	
2	Deleted			
3	All top-level pads must be listed as a vcc, vss, or io nets in the DRC/LU form. Note 1: Pads marked with "probe-only" text.dg label are exempt from this rule Note 2: pad.dg inside padvia does not apply to this check		**DRC**	
4	Minimum width of non-ESD PFET or NFET metalically connected to separate bond pads (i.e. source = metalically to pad1 and drain = metalically to pad2) Note: ESD resistors break the path between pad and diffusion marked as areaid.sigPadDiff	2000 µm	**DRC**	
5	For SEL immunity, all source and body connections must be shorted locally if the source and body are at the same DC potential. (Shorted locally implies source/body connected at the FET in metal - no separate routes for source and body). Any anomalous high resistance values (above 1000 Ω) need to be corrected via layout. All not intended-to-use body bias circuitry source and body connections must be shorted locally. Initial and date here on completion of this activity.		**NDRC** ______	______
6	Any IP with no substrate tap (compliant to substrate isolation methodology) needs to have areaid.inj layer and placed at least 100 µm away from signal pad connected diffusion		**NDRC**	
7	Any row of IOs need to end with a power pad or with an end cap cell which consists of a substrate tap and NW guard bars		**NDRC**	

	Section 12: Areaid Rules	Value	Design Signature Initials	Date
12	**Areaid Rules**			
12.1a	areaid.sigPadDiff cannot straddle nsrcdrn, psrcdrn, n+tap, or p+tap		**DRC**	
12.1b	areaid.sigPadWell cannot straddle nwell		**DRC**	
12.1c	areaid.sigPadMetNtr cannot straddle nwell, nsrcdrn, psrcdrn, n+tap, or p+tap		**DRC**	
12.2a	nsrcdrn, psrcdrn, n+tap, or p+tap connected to a signal pad (ioNet), is not inside areaid.sigPadDiff	**Recommended**	**DRC**	
12.2b	Nwell connected to signal pad (ioNet), is not inside areaid.sigPadWell	**Recommended**	**DRC**	

APPENDIX A

The approved ESD cells are located in:

DDC: s8_esd

Library: s8_esd

Use “asym” devices for products with nominal Vcc ≤3.3V and “sym” devices for products with nominal Vcc > 3.3V.

Table 1: Pin Table for All Main (primary) ESD Clamps

Category	Cell Name	Pins	Layers
pad2gnd (signal) ESD clamp / NMOS Output driver	s8_esd_signal_50_sym_hv_4k	pad,gnd,gate<0:19>, resgate,nwellRing	Met1
	s8_esd_signal_50_sym_hv_4k_dnwl_aup1	pad,gnd,gate<0:14>, resgate,<0:9>,nwellRing	Met1
	s8_esd_signal_50_sym_hv_2k	pad,gnd,gate<0:9>, resgate,nwellRing	Met1
	s8_esd_signal_40_sym_hv_2k	pad,gnd,gate<0:13>, resgate,nwellRing	Met1
	s8_esd_signal_23_sym_hv_2k	pad,gnd,gate<0:21>, resgate,nwellRing	Met1
	s8_esd_signal_30_sym_hv_2k	pad,gnd, gate<0:17>, resgate,nwellRing	Met1
	s8_esd_signal_50_sym_hv_2k_dnwl_aup1	pad,gnd,gate<0:4>, extra_gate<0:4>, resgate,nwellRing	Met1
	s8_esd_signal_50_sym_hv_2k_dnwl_aup1_b	pad, gate<9:0>, gnd, nwellRing	Met1
	s8_esd_signal_50_sym_hv_2k_dnwl_aup1_c	pad, gate<9:0>, gnd, nwellRing	Met1
	s8_esd_signal_40_sym_hv_2k_dnwl_aup1	pad, gate<9:0>, gnd, nwellRing	Met1
	s8_esd_signal_40_sym_hv_2k_dnwl_aup1_b	nwellRing, body, g<5:0>, pad<4:0>	Met1
	s8_esd_signal_40_sym_hv_2k_dnwl_aup1_c	nwellRing, body, g<5:0>, pad<4:0>	Met1
	s8_esd_signal_23_asym_hv_2k_dnwl_aup1	pad,gnd,gate<0:11>,vb nwellRing	Met1
	s8_esd_signal_23_sym_hv_2k_dnwl_aup1	pad,gnd,gate<0:11>,vb nwellRing	Met1
	s8_esd_signal_23_sym_hv_2k_dnwl_aup1_a	pad,gnd,gate<0:11>,vb nwellRing	Met1

Category	Cell Name	Pins	Layers
	s8_esd_signal_sym_nhvnative	gate<0:11> vsub, nwellring, body, drain<0:6>, source<0:5>	Poly Met1
Sonos Vpp ESD clamp	TBD	TBD	TBD
VCSEL pad ESD clamp	s8_esd_pwr2gnd_rc_50_asym_hv_4k	pwr,gnd, source	Met1
gnd2gnd ESD diode	s8_esd_gnd2gnd_120x2_lv (Not Recommended)	vssn vssi	Met1 Met1
	s8_esd_gnd2gnd_120x2_lv_isosub	vssn, vssi, vsub	Met1
LV pwr2pwr ESD clamp	s8_esd_pwr2pwr_40_1p3_1p3_lv_2k	pwr1,pwr2, gnd	Met 1
LV pwr2pwr ESD clamp (continued)	s8_esd_pwr2pwr_40_1p3_1p3_lv_2k_b	pwr1,pwr2, gnd	Met 1 LI
	s8_esd_pwr2pwr_40_1p3_1p3_lv_2k_dnwl	pwr1,pwr2, gnd nwellRing	Met 1 Li
HV pwr2pwr ESD clamp	s8_esd_pwr2pwr_50_sym_hv_2k	pwr1,pwr2,gnd	Met1
	s8_esd_pwr2pwr_40_sym_hv_2k	pwr1,pwr2,gnd	Met1
	s8_esd_pwr2pwr_21_sym_hv_2k	pwr1,pwr2,gnd	Met1
	s8_esd_pwr2pwr_40_sym_hv_2k_aup1	pwr1,pwr2, gnd	Met1 Li
	s8_esd_pwr2pwr_50_sym_hv_2k_aup1	pwr1,pwr2, gnd	Met1 Li
	s8_esd_pwr2pwr_40_1p3_1p3_hv_2k	pwr1,pwr2, gnd	Met1
Source-follower HV pwr2pwr ESD clamp	s8_esd_source_follower	sub, vpwr, vout, a	Met1
	S8_esd_stdby_pump_cell	sub, vpwr, vout_digital, a, res	Met1
pwr2gnd ESD clamp	s8_esd_pwr2gnd_rc_50_sym_hv_4k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_4k_aup1 (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k (Not recommended)	pwr,gnd, source	Met1

Category	Cell Name	Pins	Layers
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_revA (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_revB_SouthEastWest (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_dnwl (Not recommended)	pwr, gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_dnwl_revA (PSoC 3/5 Only)	pwr, gnd, source, nwellRing	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1 (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_revB_south (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_dnwl_aup	pwr,gnd, source nwellRing	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_b (Not recommended)	pwr, gnd vsub	Met1 Li
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_b_revA (PSoC 3/5 Only)	pwr, gnd , vsub gnd	Met1 Li
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_b_revB_north (PSoC 3/5 Only)	pwr, gnd , vsub gnd	Met1 Li
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_c (Not recommended)	pwr, gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_c_revB_south (PSoC 3/5 Only)	pwr, gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k_revA (PSoC 3/5 Only)	pwr,gnd,	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k_revB (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k_dnwl_aup	pwr,gnd, source nwellRing	Met1
	s8_esd_pwr2gnd_rc_21_sym_hv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_21_asym_hv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_50_casc_sym_hv_2k	pwr,pwr1,pwr2gnd	Met1

Category	Cell Name	Pins	Layers
	s8_esd_pwr2gnd_rc_40_asym_lv_4k (Not recommended)	pwr,gnd	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl_aup	pwr,gnd, source nwellRing	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revA (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revA_InsideCore (PSoC 3/5 Only)	pwr,gnd,	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revB_north (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revA_west (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl_revA (PSoC 3/5 Only)	pwr,gnd, source, nwellRing	Met1
	s8_esd_pwr2gnd_50_casc_sym_hv_2k_dnwl	pwr,pwr1,pwr2,gnd, nwellRing	Met1
	s8_esd_pwr2gnd_50_casc_sym_hv_2k_dnwl_aup1	pwr,pwr1,pwr2 gnd,nwellRing	Met1 Li
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_aup1 (Not recommended)	pwr, gnd, vsub	Met1 Li
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_aup1_revA (PSoC3/5 Only)	pwr, gnd, vsub gnd	Met1 Li
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl_aup1 (Not recommended)	pwr, gnd, vsub, nwellRing	Met1 Li

Table 2: Pin Table for Regulator Blocks

Category	Cell Name	Pins	Layers
Regulator Block	mtdr_io_reg_mockup	vpwr_out	Met2
	s8tee_reg_top	vpwr_io	Met2

Note: The Information in the above table is not a part of s8_esd library..

Table 2a: Table for PMOS Output Driver Blocks

Category	Cell Name
PMOS Output Driver	pEsdFet

Note: The Information in the above table is not a part of s8_esd library. The cell comes from s8_esd_dv library.

Table 3: Pin Table for NFET Local Clamps

Category	Cell Name	Pins	Layers
Local ESD clamp	s8_esd_signal_5_sym_hv_local	in,nbody,vgnd,nwellRing	Met1
	s8_esd_signal_5_sym_hv_local_5term	in,nbody,vgnd,gate, nwellRing	Met1
	s8_esd_signal_5_sym_hv_local_5term_dnwl	in,nbody,vgnd,gate, nwellRing	Met1
	s8_esd_signal_5_sym_lv_local	in,nbody,vgnd,nwellRing	Met1

Table 4: Pin Table for NFET Local Diode

Category	Cell Name	Pins	Layers
Local ESD diode	s8_esd_localdiode_hv	vgnd,nwell,p	Met1
	s8_esd_localdiode_lv	vgnd,nwell,p	Met1

Table 5: Pin Table for Ground to Ground Clamps

Category	Cell Name	Pins	Layers
gnd2gnd ESD diode	s8_esd_gnd2gnd_120x2_lv (Not recommended)	vssn vssi	Met1 Met1
	s8_esd_gnd2gnd_120x2_lv_isosub	vssn, vssi, vsub	Met1

Table 6: Pin Table for Low Voltage Power to Power ESD Clamps

Category	Cell Name	Pins	Layers
LV pwr2pwr ESD clamp	s8_esd_pwr2pwr_40_1p3_1p3_lv_2k	pwr1,pwr2, gnd	Met1
LV pwr2pwr ESD clamp	s8_esd_pwr2pwr_40_1p3_1p3_lv_2k_b	pwr1,pwr2, gnd	Met 1 LI
	s8_esd_pwr2pwr_40_1p3_1p3_lv_2k_dnwl	pwr1,pwr2, gnd nwellRing	Met1 Li

Table 7: Pin Table for Power to Power ESD Clamps

Category	Cell Name	Pins	Layers
HV pwr2pwr ESD clamp	s8_esd_pwr2pwr_50_sym_hv_2k	pwr1,pwr2,gnd	Met1
	s8_esd_pwr2pwr_40_sym_hv_2k	pwr1,pwr2,gnd	Met1
	s8_esd_pwr2pwr_21_sym_hv_2k	pwr1,pwr2,gnd	Met1
	s8_esd_pwr2pwr_40_sym_hv_2k_aup1	pwr1,pwr2, gnd	Met1 Li
	s8_esd_pwr2pwr_50_sym_hv_2k_aup1 s8_esd_pwr2pwr_40_1p3_1p3_hv_2k	pwr1,pwr2, gnd pwr1,pwr2, gnd	Met1 Li Met1

Table 8: Pin Table for Power to Ground ESD Clamps

Category	Cell Name	Pins	Layers
pwr2gnd ESD clamp	s8_esd_pwr2gnd_rc_50_sym_hv_4k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_4k_aup1 (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_revA (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_revB_SouthEastWest (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_dnwl (Not recommended)	pwr, gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_dnwl_revA (PSoC 3/5 Only)	pwr, gnd, source, nwellRing	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1 (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_revB_south (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_dnwl_aup	pwr,gnd, source nwellRing	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_b (Not recommended)	pwr, gnd vsub	Met1 Li
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_b_revA (PSoC 3/5 Only)	pwr, gnd , vsub gnd	Met1 Li
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_b_revB_north (PSoC 3/5 Only)	pwr, gnd , vsub gnd	Met1 Li
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_c (Not recommended)	pwr, gnd, source	Met1
	s8_esd_pwr2gnd_rc_50_sym_hv_2k_aup1_c_revB_south (PSoC 3/5 Only)	pwr, gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k_revA (PSoC 3/5 Only)	pwr,gnd,	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k_revB (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_sym_hv_2k_dnwl_aup	pwr,gnd, source nwellRing	Met1

Category	Cell Name	Pins	Layers
	s8_esd_pwr2gnd_rc_21_sym_hv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_21_asym_hv_2k (Not recommended)	pwr,gnd, source	Met1
Category	**Cell Name**	**Pins**	**Layers**
pwr2gnd ESD clamp	s8_esd_pwr2gnd_50_casc_sym_hv_2k	pwr,pwr1,pwr2gnd	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_4k (Not recommended)	pwr,gnd	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl_aup	pwr,gnd, source nwellRing	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revA (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revA_InsideCore (PSoC 3/5 Only)	pwr,gnd,	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revB_north (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_revA_west (PSoC 3/5 Only)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl (Not recommended)	pwr,gnd, source	Met1
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl_revA (PSoC 3/5 Only)	pwr,gnd, source, nwellRing	Met1
	s8_esd_pwr2gnd_50_casc_sym_hv_2k_dnwl	pwr,pwr1,pwr2,gnd, nwellRing	Met1
	s8_esd_pwr2gnd_50_casc_sym_hv_2k_dnwl_aup1	pwr,pwr1,pwr2 gnd,nwellRing	Met1 Li
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_aup1 (Not recommended)	pwr, gnd, vsub	Met1 Li
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_aup1_revA (PSoC3/5 Only)	pwr, gnd, vsub gnd	Met1 Li
	s8_esd_pwr2gnd_rc_40_asym_lv_2k_dnwl_aup1 (Not recommended)	pwr, gnd, vsub, nwellRing	Met1 Li

Table 9: Pin Table for ESD Poly Resistors

Category	Cell Name	Pins	Layers
ESD resistor	s8_esd_res75only	rout,pad	Met1
	s8_esd_res75only_small	rout,pad	Met1
	s8_esd_res75only_noshorts	rout, pad	Met1
	s8_esd_res75only_noshorts_nometal	pad, rout	Li
	s8_esd_res250	rout,cut1a,cut1b,cut2a,cut2b,cut3a,cut3b,cut4a,cut4b,pad	Met1
	s8_esd_res250only s8_esd_res250only_small	rout,pad rout, pad	Met1 Met1

Table 10: Pin Table for Pad-Gnd ESD Diodes

Category	Cell Name	Pins	Layers
pad2gnd ESD and RF pad current steering diodes	s8_esd_paddiode2gnd_100_hv	p,n,nwl	Met1
	s8_esd_paddiode2gnd_200_hv	p,n,nwl	Met1
	s8_esd_paddiode2gnd_300_hv	p,n,nwl	Met1
	s8_esd_paddiode2gnd_100_dnwl_hv	p,n,nwl	Met1
	s8_esd_paddiode2gnd_200_dnwl_hv	p,n,nwl	Met1
	s8_esd_paddiode2gnd_300_dnwl_hv	p,n,nwl	Met1

Table 11: Pin Table for Pad-Pwr ESD Diodes

Category	Cell Name	Pins	Layers
pad2pwr, RF pad current steering, and VCSEL pad diodes	s8_esd_paddiode2pwr_100_hv	n,p,sub	Met1
	s8_esd_paddiode2pwr_200_hv	n,p,sub	Met1
	s8_esd_paddiode2pwr_300_hv	n,p,sub	Met1

Table 12: Pin Table for ESDRB Approved Cells

Category	Cell Name	Pins	Layers
VCSEL Driver Pull-down	s8_esd_vcseldrv_casc_nmos	Vnb nCascGate, nMainGate vgnd, vcsel_rtn	Li Poly Met1
VCSEL Driver Pull-up	s8_esd_vcseldrv_casc_pmos	Vpb pCascGate, pMainGate vpwr, vcsel_out	Li Poly Met1
VCSEL Driver Pull-down	s8_esd_vcseldrv_casc_nmos_no_power	Vnb nCascGate, nMainGate vgnd, vcsel_rtn	Li Poly Met1
VCSEL Driver Pull-up	s8_esd_vcseldrv_casc_pmos_no_power	Vpb pCascGate, pMainGate vpwr, vcsel_out	Li Poly Met1
Nhvnative Driver pull-up	s8psoc3io_sio_pudrvr_reg_pu_natives	vnb, vpb_drvr pad, vgnd_io, vref_nng, drvhi_h, vref_int, d1, d2, d3, d4, d5	Met1 Met2
Nhvnative Driver pull-up	s8ppscio_sio_pudrvr_reg_pu_natives	vnb, vpb_drvr pad, vgnd_io, vref_nng, drvhi_h, vref_int, d1, d2, d3, d4, d5	Met1 Met2

APPENDIX B – S8 LU DRC Form Description

Verification Form Entry	Description	Entry Format	Example
Latchup Switches: local_sub	Assumes an isolated substrate where areaid:substrateCut is used		
Latchup Switches: block_level	Used for IP-level checks. IO nets are assigned to pad-level checks		
Select Check	Checks only the specified rule and reports if that rule is violated	Specify in the batch options of the GUI or commandline -selectCheck rulename. Rule # is from the error report	
Joined Named Nets	Ties all like-named nets together		
Save Nets File	Saves a copy of the current nets list		
VSS Nets	List of all ground nets	Space delimited. Regulated grounds in [ ]	vss1 vss2 [vgnd]
VCC Nets	List of all power nets	Space delimited. Regulated powers in [ ]	vcc1 vcc2 [vpwr]
I/O Pad Nets	List of all I/O (signal) nets, including inputs, outputs and I/Os	Space delimited	a1 a2 io3 oeb
Nets File	Imports previously saved nets list file	Specify nets list path in the nets file box	
Inp Pad Nets(5V)	5V only input nets	Space delimited	a1 a2 a3 ceb
Output or IO Pad Nets (5V)	All output and I/O nets	Space delimited	io1 io2 outa
Vpp Pad	List of Vpp nets for SONOS programming pad(s)	Space delimited	vpp
Esd 7_6Nets	Refers to MTDR section 7, rule 6 :Specifies signal pads that may be held at a DC level of anywhere between half of Vcc to Vcc inclusive for any time longer than the inverse of the smallest AC signal pad frequency of the product,	Space delimited	io1 io2
Esd 8_4Nets	Refers to MTDR section 8, rule 4 = pwr2pwr clamps required for signal transitions across power domains	Space delimited	
Main Ground Supply	Ground supply with the highest capacitance (typically substrate)	Regulated grounds in [ ]	[vgnd]
Main Power Supply	Power supply with the highest capacitance	Regulated supply in [ ]	[vpwr]
Lv Power Supply	Power supply connected to LV components	Space delimited. Regulated powers NOT in [ ]	vpwra vpwr
Hv Power Supply (3.3V or 5V)	Power supply connected to HV components	Space delimited. Regulated powers NOT in [ ]	vcca vpgm
Power Ground Pairs	Power nets and their associated ground nets	Pairs in ( ), space delimited	(vcc1 vss1) ([vpwr] [vgnd])
VCSEL driver Pairs	VCSEL net names	Space delimited	vscel_out1 vcsel_out2
Short Nets	Allows shorting of series resistances, allowing rule checks of resistively connected nets	Shorted nets separated by ":"	res_in:res_out res_in:inp